Secretary-Style Split Search for Decision Trees

Mathias Valla[a]

[a] *Chaire DIALog, Institut Louis Bachelier, Paris, France*

Abstract

The standard CART splitter examines all admissible covariate–threshold pairs at each node, which can make tree induction computationally expensive when many continuous features are available. This paper asks whether such exhaustive split enumeration is always necessary to grow accurate decision trees. We study a small family of secretary-style split-search rules that evaluate only a random subset of candidate splits and stop early according to simple sequential criteria. The proposed strategies include a single-stream secretary rule over random splits, a double secretary rule over covariates and thresholds, a covariate-level secretary rule with exhaustive within-covariate refinement, and a parametric secretary rule calibrated from exact gain evaluations. We also adapt the block-rank algorithm as a rank-adaptive secretary-style alternative.

The methods are compared with the standard exhaustive `scikit-learn` splitter on a broad collection of regression and classification datasets from the PMLB benchmark suite, using repeated runs to quantify both average behavior and stochastic variability. In regression, aggressive secretary rules achieve substantial speedups but usually at too high a predictive cost. In classification, conservative and rank-adaptive variants, especially covariate-level secretary with exhaustive refinement and the block-rank rule, provide meaningful training-time reductions with comparatively modest losses. The parametric secretary strategy is not competitive in practice because its fitting overhead outweighs the benefit of early stopping. Secretary-style split search is therefore useful only under sufficiently conservative stopping rules and primarily in favorable classification settings.

Keywords: decision trees, CART, secretary problem, randomized split search, benchmark evaluation, computational efficiency

1. Introduction

Classification and regression trees remain widely used for tabular data because they combine flexibility, interpretability, and strong empirical performance [1]. Their standard induction rule is greedy: at each internal node, CART evaluates candidate covariate–threshold pairs and selects the split maximizing the impurity decrease. For continuous covariates, the number of admissible thresholds can be large, so exhaustive split search may dominate training time.

This raises a basic question: is it always necessary to evaluate all admissible splits in order to grow an accurate tree? In many nodes, only a small subset of candidates attains gains close to the optimum. Existing randomized tree procedures exploit this idea indirectly. Random forests subsample covariates [2], and extremely randomized trees randomize threshold choice [3]. These methods, however, are not formulated as stopping rules on a stream of candidate gains.

We recast split search as a secretary-style online selection problem. In the classical secretary problem, items arrive sequentially in random order and the decision maker must stop online to select a highly ranked item [4, 5]. The classical skip rule, with asymptotic success probability $1/e$, shows that a short exploration phase can be followed by a simple acceptance rule with a nontrivial guarantee [4, 5]. More recent work has introduced rank-adaptive rules based on the availability of a single sample, including the block-rank family [6]. These ideas suggest a natural analogue for tree induction: if covariates and thresholds are explored in random order, split selection becomes a stopping problem on a stream of exact impurity gains.

We study four secretary-style split-search strategies for CART, ranging from direct split-level stopping to covariate-level stopping and model-based calibration, and also include a rank-adaptive block-rank variant and a one-sample prophet-style baseline.

The contribution is empirical and methodological rather than theoretical. We benchmark the proposed rules against the standard exhaustive `scikit-learn` splitter on a large collection of regression and classification datasets from the PMLB benchmark suite [7]. The goal is to characterize the time–accuracy trade-off, quantify the variability induced by randomized split exploration, and identify which stopping rules offer the best practical compromise. An open-source implementation of all methods and the scripts used to reproduce the experiments are released publicly.[1]

2. Methods

2.1. Split-search setting

Consider a node $\mathcal{N}$ containing n observations $\{(X_i, Y_i)\}_{i=1}^n$ with $X_i \in \mathbb{R}^p$. Let $\mathcal{J} = \{1, \ldots, p\}$ be the set of covariates, and let $\mathcal{S}_j$ be the set of admissible thresholds for covariate j. A candidate split is a pair (j, s) with $s \in \mathcal{S}_j$, producing children of sizes $(n_{\mathrm{L}}, n_{\mathrm{R}})$. For an impurity I, the gain of split (j, s) is

$$G_{j,s} = I(\mathcal{N}) - \frac{n_{\mathrm{L}}}{n} I(\mathcal{N}_L) - \frac{n_{\mathrm{R}}}{n} I(\mathcal{N}_R). \tag{1}$$

For regression, I is the mean squared error impurity. For classification, we consider Gini and entropy impurities [1]. Once a threshold is sampled, its gain is computed exactly on the full node data. Randomness therefore comes only from the order in which covariates are visited and from random sampling of thresholds within covariates.

To study early stopping, we impose a random order on covariates and, within each visited covariate, a random order on thresholds. Split search then becomes a sequential decision problem on a stream of exact gains. In the experiments, the exploration budget at a node is parameterized as a function of node size n, with schedules of order n/e, $0.1n$, $\log n$, and $\sqrt{n}$.

2.2. Stopping rules

Method S treats split search as a single secretary problem over a random stream of candidate pairs (j, s). Given a budget B, it rejects the first r candidates and then accepts the first subsequent split whose gain exceeds all gains observed in the exploration phase. If no such split appears, the best candidate seen so far is returned. This is the direct adaptation of the classical secretary rule [4, 5].

Method S_{all} applies the secretary rule at the covariate level. For a visited covariate j, its score is

$$V_j = \max_{s \in \mathcal{S}_j} G_{j,s}.$$

[1] Code available at https://github.com/MathiasValla/EarlyStoppingTrees

Covariates are processed in random order, and a secretary rule is applied to the sequence (V_j). When a covariate is accepted, the best threshold within that covariate is returned. This method is more expensive than S for each inspected covariate, but it preserves exact within-covariate optimization.

Method S^2 introduces a nested secretary structure. The outer rule is applied to covariates in random order. For each visited covariate j, an inner secretary rule is applied to a random threshold stream inside $\mathcal{S}_j$. The inner rule returns a threshold $\hat{s}_j$ and a realized gain $\tilde{V}_j = G_{j,\hat{s}_j}$, and the outer rule is then applied to the sequence $(\tilde{V}_j)$. The outer stage selects the covariate and the inner stage selects the threshold.

Method S_{par} is a model-based variant. For a given covariate j, a small number of random thresholds is first evaluated exactly, yielding gain samples used to fit a working distributional model. A stopping threshold is then derived from this fitted distribution, for example through a high quantile, and thresholds are sampled until one exceeds the fitted acceptance level. The goal is not to claim an exact generative law for random-threshold gains, but to provide a calibrated stopping rule based on a small number of exact evaluations.

The block-rank algorithm replaces the single exploration threshold of the classical secretary rule by a sequence of rank-based thresholds that become more permissive across successive budget blocks [6]. In the present context, this yields a rank-adaptive stopping rule on a random stream of split gains. We evaluate it as a structured alternative to the simple secretary baseline.

We also include a one-sample prophet-style baseline. For each covariate, one random threshold is sampled and evaluated exactly, and the maximum of these gains is used as an empirical acceptance threshold. A second pass then considers new random split candidates in random order and accepts the first one whose gain exceeds this threshold. This baseline is motivated by prophet inequalities from samples [8, 9].

2.3. Essential theory

The theory is used only to motivate the stopping rules and the parametric calibration. We do not attempt a full analysis of tree induction under secretary-style exploration.

The classical secretary problem considers N distinct items arriving in uniformly random order. If the first r items are rejected and the first subsequent record is accepted, then the probability of selecting the maximum is

$$\frac{r}{N}\sum_{i=r+1}^{N}\frac{1}{i-1},$$

which is asymptotically maximized by $r \sim N/e$, yielding success probability $1/e$ [4, 5]. This benchmark motivates the exploration size N/e.

Under the random-permutation model, the skip-r secretary rule is asymptotically optimal for $r \sim N/e$, and its success probability converges to $1/e$ as $N \to \infty$.

For regression trees with MSE impurity, the gain of split (j, s) can be written exactly as

$$G_{j,s} = \frac{n_{\mathrm{L}}n_{\mathrm{R}}}{n^2}(\bar{Y}_L - \bar{Y}_R)^2. \tag{2}$$

Assume

$$Y_i = \mu_i + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

and let $\Delta_{j,s} = \mu_L - \mu_R$ be the population mean gap induced by the split. Then

$$G_{j,s} \overset{d}{=} \frac{\sigma^2}{n}\chi_1^2(\lambda_{j,s}), \qquad \lambda_{j,s} = \frac{\Delta_{j,s}^2\, n_{\mathrm{L}}n_{\mathrm{R}}}{\sigma^2 n}. \tag{3}$$

This is exact. Under the null split, $\lambda_{j,s} = 0$, so the gain is a scaled central chi-square variable.

Under Gaussian noise, the MSE gain of a fixed split has an exact scaled noncentral chi-square distribution with one degree of freedom.

For classification, exact finite-sample gain laws are less tractable, so we use standard asymptotic approximations. For entropy gain, $2n$ times the gain is the likelihood-ratio statistic for testing independence between the class label and the child indicator; under local alternatives it is asymptotically noncentral chi-square with $K - 1$ degrees of freedom. For Gini gain, the identity

$$G_{j,s} = \pi(1-\pi)\|\hat{p}_L - \hat{p}_R\|_2^2, \qquad \pi = \frac{n_{\mathrm{L}}}{n},$$

shows that the gain is quadratic in the difference of empirical child class proportions. In binary classification this leads under local alternatives to

Table 1: Summary of the PMLB benchmark used in the study.

Task	N	Median n	Median p	$n_{\min}$–$n_{\max}$	$p_{\min}$–$p_{\max}$
Regression	107	500	10	47–40768	2–124
Classification (Gini)	120	736	13	32–58000	2–240
Classification (Entropy)	119	675	13	32–58000	2–216

a scaled noncentral chi-square approximation, while in multiclass settings weighted chi-square or gamma approximations are more appropriate.

If gain samples are modeled as $G \sim a\chi^2_k$, then G belongs to a gamma family. If $G_1, \ldots, G_q$ are i.i.d. samples from this model, the sample mean $\bar{G} = q^{-1} \sum_{\ell=1}^{q} G_\ell$ is gamma-distributed, and the natural moment estimator of the scale parameter is $\hat{a} = \bar{G}/k$. Standard chi-square concentration inequalities, such as Laurent–Massart bounds, then justify conservative fitted quantiles for S_{par} [10].

All methods were implemented in a common open-source package built around a modified tree splitter, with identical tree-growing hyperparameters across methods. This ensures that the observed differences are attributable to split-search rules rather than to other changes in the induction procedure.

3. Results

3.1. Experimental protocol

We compared secretary-style split search against the standard exhaustive splitter of `scikit-learn` on a broad collection of regression and classification datasets from the PMLB benchmark suite [7]. For each dataset, preprocessing, train/test splitting, and tree hyperparameters were held fixed across methods. Because the proposed procedures are randomized, each dataset–method pair was evaluated over 100 independent runs.

The primary computational metric is wall-clock training time. Predictive performance is measured by RMSE in regression and by F1-score in classification; accuracy and additional visualizations are reported in the supplementary material. Results are normalized relative to the exhaustive splitter through a speedup factor and a relative loss metric.

3.2. Global trade-off

Figure 1 summarizes the main Pareto comparison between speedup and predictive degradation. Each point corresponds to one dataset summarized by the median over the 100 runs. Size-stratified versions of the same Pareto comparison are reported in Supplementary Figure S1. Three conclusions are immediate: all secretary-style rules save computation; the time–accuracy trade-off depends strongly on the task; and covariate-level and rank-adaptive rules are systematically safer than aggressive split-level secretary rules.

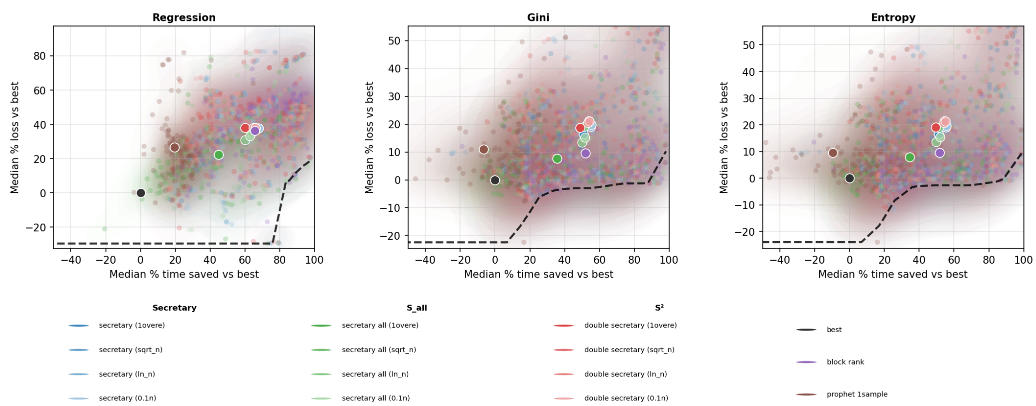

Figure 1: Global Pareto comparison of secretary-style split-search methods. Each point represents one dataset summarized by the median over 100 runs. The horizontal axis reports the median percentage of training time saved relative to the exhaustive `scikit-learn` splitter, and the vertical axis reports the median percentage loss relative to the same baseline. Left: regression with RMSE loss. Middle: classification with Gini impurity and F1 loss. Right: classification with entropy impurity and F1 loss. Large white-edged markers indicate the cross-dataset medians of the corresponding methods.

Table 2 reports the benchmark summary using the theoretically motivated n/e schedule as the default representative for the secretary families. The table reports median speedup, median loss, 90th percentile loss, and two practical probabilities: the fraction of datasets for which the median loss remains below 5%, and the fraction for which the median speedup is at least 20%.

The regression results are unfavorable to secretary-style approximation. Simple secretary and double secretary achieve median speedups between 2.65× and 3.42×, but their median RMSE losses exceed 40%. The block-rank rule is only slightly less damaging, with median loss 38.7%. The least harmful

Table 2: Global summary by method. For the secretary families, the n/e schedule is used as the default representative. Speedup is relative to the exhaustive `scikit-learn` splitter. Loss is expressed in percent. Full results for all schedules are reported in the supplementary material.

Task	Method	Median speedup	Median loss (%)	90th pct. loss (%)	$P(\text{loss} \leq 5\%)$	$P(\text{speedup} \geq 20\%)$
Regression	Exhaustive	1.00	0.0	0.0	1.000	0.000
Regression	S	3.42	41.9	58.0	0.086	1.000
Regression	S^2	2.65	42.2	59.7	0.068	0.990
Regression	S_{all}	1.80	22.6	43.5	0.184	0.854
Regression	Block-rank	3.30	38.7	56.3	0.048	1.000
Regression	1-sample prophet	1.27	23.9	57.1	0.095	0.724
Classification (Gini)	Exhaustive	1.00	0.0	0.0	1.000	0.000
Classification (Gini)	S	1.88	11.3	47.7	0.381	0.873
Classification (Gini)	S^2	1.67	12.6	52.9	0.343	0.833
Classification (Gini)	S_{all}	1.40	3.8	23.5	0.593	0.620
Classification (Gini)	Block-rank	1.85	5.2	24.7	0.500	0.958
Classification (Gini)	1-sample prophet	1.12	7.5	26.3	0.424	0.288
Classification (Entropy)	Exhaustive	1.00	0.0	0.0	1.000	0.000
Classification (Entropy)	S	1.87	11.2	48.3	0.373	0.890
Classification (Entropy)	S^2	1.72	12.7	53.3	0.339	0.853
Classification (Entropy)	S_{all}	1.41	4.9	25.3	0.523	0.633
Classification (Entropy)	Block-rank	1.83	5.0	25.9	0.500	0.966
Classification (Entropy)	1-sample prophet	1.09	6.0	24.3	0.475	0.271

variants are S_{all} and the one-sample prophet baseline, whose median losses remain around 23%, but their speedups are much smaller.

The classification results are more favorable. With both Gini and entropy impurities, the strongest compromises are obtained by S_{all} and block-rank. For Gini, S_{all} achieves median speedup 1.40× for median loss 3.8%, whereas block-rank reaches 1.85× for median loss 5.2%. For entropy, the corresponding values are 1.41× and 4.9% for S_{all}, and 1.83× and 5.0% for block-rank. By contrast, simple and double secretary are somewhat faster but substantially less reliable, with median losses between 11% and 13%. The one-sample prophet baseline is safer than S and S^2, but its speedups are modest.

A plausible explanation for the strong task asymmetry observed in the benchmark is that classification impurities are more tolerant to approximate split search than regression impurity. For Gini and entropy, many nearby thresholds can induce very similar child class proportions and therefore nearly equivalent impurity reductions. In such cases, partial random exploration may still identify a split whose downstream predictive effect is close to that of the exhaustive optimum. By contrast, regression gains based on MSE appear to depend more sharply on the precise threshold location, because small changes in the partition can alter child means and variances continuously and hence affect the final predictions more directly. This suggests that the gain landscape over thresholds is typically flatter in classification and more localized in regression, making secretary-style early stopping substantially safer in the former setting than in the latter.

3.3. Reliability and dataset structure

Figure 2 complements the median-based summaries by showing reliability under explicit operating constraints. Aggressive methods such as S and S^2 can satisfy moderate loss tolerances on a non-negligible subset of runs, especially in classification, but they also exhibit heavier tails of poor outcomes. Size-stratified reliability curves are reported in Supplementary Figure S6, and the full grid of joint success probabilities over all datasets is reported in Supplementary Figure S7.

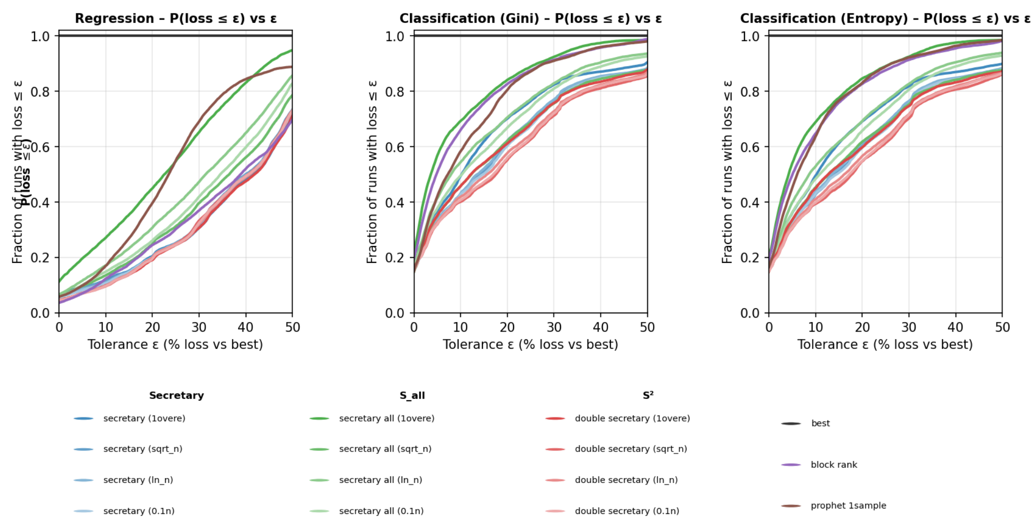

Figure 2: Reliability summaries for secretary-style split-search methods. Top row: empirical probability of achieving a predictive loss below a tolerance ε. Bottom panels: empirical probability of simultaneously achieving at least a prescribed time saving and a predictive loss below a prescribed tolerance. Separate panels are shown for regression, classification with Gini impurity, and classification with entropy impurity. Size-stratified and method-specific variants are reported in the supplementary material.

By contrast, S_{all} and block-rank are less aggressive but more reliable. This explains why simple and double secretary can appear competitive under permissive thresholds while still performing poorly in dataset-level medians. Their occasional strong runs do not compensate for their broader and more skewed loss distributions. Supplementary Figures S3–S5 report task-specific ridgeline summaries of within-dataset variability, and Supplementary Figures S9–S11 provide complementary whisker summaries of the same repeated-run variability.

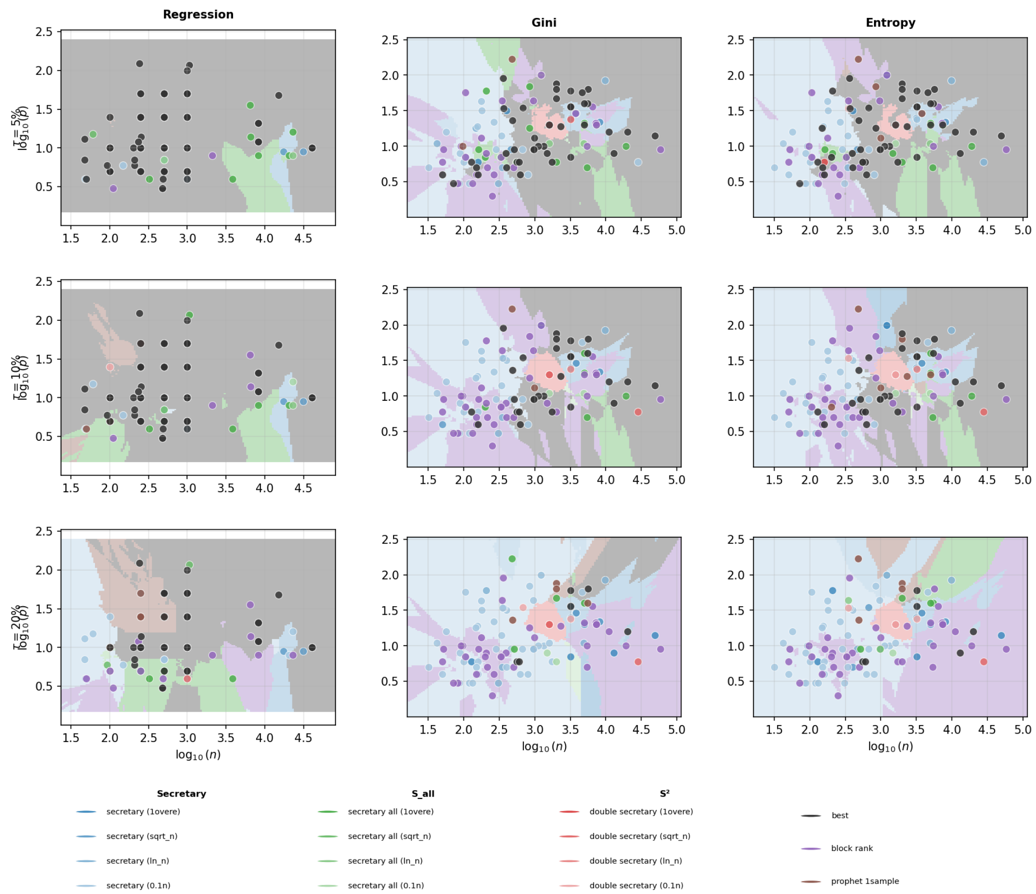

Figure 3: Regime maps as a function of dataset size and dimensionality. Each point is a benchmark dataset represented in the $(\log_{10} n, \log_{10} p)$ plane and colored by the split-search strategy that gives the best practical trade-off under a fixed speedup–loss criterion. Columns correspond to loss tolerances $\tau = 5\%, 10\%, 20\%$. Rows correspond to regression, classification with Gini impurity, and classification with entropy impurity.

Figure 3 shows that the usefulness of secretary-style split search depends on dataset structure. Classification tasks are clearly more favorable than regression tasks. Within classification, and under the chosen speedup–loss criterion, covariate-level and rank-adaptive methods dominate larger portions of the regime map than split-level secretary rules. In regression, exhaustive split search remains the safest choice on much of the benchmark, and secretary-style rules are only competitive in limited regions.

3.4. Sensitivity and negative result for the parametric strategy

We examined four exploration schedules based on node size: n/e, $0.1n$, $\log n$, and $\sqrt{n}$. The full results are reported in the supplementary material. Changing the schedule affects the quantitative trade-off more than the qualitative ranking. For S and S^2, more aggressive schedules produce slightly larger speedups at the price of slightly worse predictive degradation. For S_{all}, the n/e schedule consistently provides the best compromise, especially in classification, which justifies its use as the default representative in the main text. The corresponding size-stratified Pareto analyses and reliability summaries are reported in Supplementary Figures S1, S6, and S7.

The parametric secretary method S_{par} was evaluated in an initial screening study over a grid of fitting sample sizes and quantile levels. Although statistically motivated by gain-distribution approximations, it was consistently dominated in practice: the cost of fitting the working model outweighed any benefit from early stopping, and the resulting trees were typically both slower and less accurate than the exhaustive baseline. For this reason, S_{par} was not retained in the main comparative benchmark, its full screening analysis is reported in Supplementary Figure S2. This negative result shows that probabilistic calibration is not useful by itself when the fitting overhead is too large relative to the gain-computation cost.

Additional regime-map and repeated-run variability analyses are provided in Supplementary Figures S8–S11.

4. Conclusion, limitations, and outlook

This paper asked whether exhaustive CART split search can be replaced, at least in part, by secretary-style sequential selection without incurring an unacceptable loss in predictive performance. The answer depends strongly on both the stopping rule and the prediction task. Secretary-style split search does reduce training time, sometimes substantially, but aggressive split-level strategies such as simple secretary and double secretary often do so at too high a predictive cost. More conservative or more structured variants remain practically competitive. In particular, the covariate-level strategy S_{all} and the rank-adaptive block-rank rule emerge as the most convincing compromises, especially in classification, where they provide meaningful speedups with comparatively modest degradation in predictive performance.

The study reveals a clear task asymmetry. Regression trees are much less tolerant to incomplete split exploration than classification trees. On the regression benchmarks considered here, even the best secretary-style variants remain far from the exhaustive splitter in predictive preservation. In classification, by contrast, the losses are often moderate enough to make the time–accuracy trade-off practically interesting. This suggests that the value of approximate split search depends on the impurity criterion and on the sensitivity of the task to local split errors.

A second important outcome is negative but informative. The parametric secretary variant S_{par}, although statistically motivated by gain-distribution approximations, was consistently unsuccessful in practice. The cost of fitting a working model dominated any computational benefit from early stopping. This cautions against assuming that a principled probabilistic calibration will automatically lead to an effective split-search procedure.

The study is limited to single trees, benchmark datasets, and a deliberately small family of stopping rules, and is intended primarily as an empirical characterization rather than a complete theory.

The most immediate extension is to study the same split-search rules in bagging and boosting. Because split-search cost is incurred at every node and across many trees, even moderate node-level savings may accumulate into substantial end-to-end gains. At the same time, ensembling may reduce the predictive effect of occasional suboptimal splits, potentially making secretary-style rules more attractive than they are in single trees. A second direction is methodological: the performance of the block-rank rule suggests that adaptive rank-based or sample-based stopping rules may be preferable to the classical secretary rule in this context. Overall, secretary-style split search is not a universal replacement for exhaustive CART search, but in favorable settings, especially in classification, it provides a simple and interpretable way to trade split-search time against predictive performance.

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